



Lesson 2: Machine learning

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In this lesson, learners will learn to describe the difference between data-driven and rule-based approaches to solving problems. They will explore how machine learning systems are created using a data-driven approach, including supervised learning.

Objectives

- Objective 1 Describe the difference between 'data-driven' and 'rule-based' approaches to application development
- Objective 2 Define the role of machine learning in creating models
- Objective 3 Describe the impact of data on the output of a machine learning (ML) model

Keywords

- Data-driven
- Machine learning
- Accuracy
- Training data

Lesson outline

Activity	Objective	Description	Time
Activity 1: The intelligent piece of paper	1	Learners see how rule-based systems can do some impressive things.	15 mins
Activity 2: Rule based vs Data-driven systems	1, 3	Learners explore how newer AI systems are data-driven and are created using lots of data.	10 mins
Activity 3: Solving problems with machine learning	2, 3	Learners discover how a data-driven approach can be used to generate images.	15 mins
Optional: Emoji predictor	1, 3	Learners use training data to make predictions about emoji reactions.	~15 mins
Exit ticket	1, 2, 3	Learners review key concepts about machine learning by completing short sentences.	5 mins

Resources

Computer with internet access

Printed resource

- Educator resource: The intelligent piece of paper

Links

- rpf.io/xai-2-v1
- rpf.io/xai-2-v2

Subject knowledge

Activity 2 and 3: These activities will introduce the term ‘machine learning’, so we recommend that you familiarise yourself with the definition below:

Machine learning (ML) is an area of artificial intelligence where data-driven models produce predictions based on patterns in large amounts of data. Machine learning systems are called ‘data-driven’ because they use a lot of examples – data – to work. This is unlike traditional programming where step-by-step instructions are written to solve a problem.

Because they are data-driven, **machine learning systems can produce useful results even for complicated problems** that are not well defined. However, for the same reason, machine learning systems can also produce less consistent results and be harder to debug than rule-based programming.

Examples of machine learning applications include tools to process speech, identify faces, recommend songs, and diagnose illnesses.

We also recommend that you watch the video on slide 8 before the lesson, so you can anticipate any questions your learners may have.

Assessment opportunities

Objective	Description
1	Activity 1: You could assess learners’ ability to describe the difference between ‘data-driven’ and ‘rule-based’ approaches by asking them questions at the end of the ‘intelligent piece of paper’ activity. Activity 2: Using the questions on the slide after the video.
2	Activity 2 and 3: Assess their understanding of the role in machine learning in creating models by using the drag and drop numbering activities.
3	Activity 3: You can assess learners’ understanding of how data impacts the output of a machine learning model by asking them how they would improve the output of a machine learning model that generates images of hands.

Lesson walkthrough

The **Success criteria** on slide 3 translate lesson objectives into actionable, student-friendly targets.

They clearly define what the students should be able to do by the end of the lesson.

Slide deck labels appear at the top of each slide to help you navigate the activities in the lesson.

This lesson includes an **optional activity** that will take around 15 minutes. If you do not want to include it, please skip slides 22–25. These slides have been hidden by default.

15 mins

Activity 1: The intelligent piece of paper

Slide 2

Resource required: Educator resource: The intelligent piece of paper

As learners enter the classroom, ask them to discuss their thoughts on the question “What is intelligence?” with the person next to them.

Educator guidance

The aim of this activity is to set the scene and to focus the learners’ attention. You do not need to take answers from the learners; instead, reveal the definition on the slide: **“People often say intelligence is the ability to learn and handle new situations.”**

Highlight that the key point is being able to learn something and use that learning to **adapt and react to new situations**.

Slide 4

Hold up the intelligent piece of paper to show to the class. The side with the instructions on should be facing you, and the class should see the blank side.

Announce that this piece of paper is intelligent. Tell the learners that not only is it intelligent, but it is also more intelligent than anyone in the room. Assure the learners that it is just a piece of paper and there is no computer chip embedded in it.

Ask the learners to raise their hands if they think they are more intelligent than the piece of paper.

Congratulate the learners who did not think they are more intelligent than the piece of paper, and tell them they are very wise. Tell the learners who think they are more intelligent than the piece of paper that you can prove that the piece of paper is intelligent.

Tell the class that to prove that the piece of paper is intelligent, they will play tic-tac-toe (noughts and crosses) against it.

Slide 5

Inform the class that the piece of paper has never lost a game of tic-tac-toe and it will not lose to them.

Pick two volunteers from the class to play the game:

- One volunteer will represent all of humanity. If possible, pick one of the learners who thought they were more intelligent than the piece of paper.
- One volunteer will act as the piece of paper’s assistant. Explain that because it is a piece of paper, with no robotic controls to be able to play by itself, it needs a volunteer to help it play. Give the piece of paper to the volunteer and they will see the instructions on the piece of paper. Instruct them to follow the instructions on the piece of paper and do exactly as it says.

Draw a tic-tac-toe grid on a whiteboard for the class to see.

Ask the volunteer with the piece of paper to read out the first instruction. Say that the piece of paper wants to go first. Comment that it is quite clever of it to want to go first. If someone complains about it being unfair, point out that the game should just end in a draw whoever goes first. Going second is not a reason to lose.

The volunteer with the piece of paper should then read out the first move and make that move: playing in a corner. The player representing humanity can then take a turn.

Allow the volunteers to continue playing the game. Support the learner with the piece of paper, ensuring that they read out the relevant instruction so that the rest of the class can hear it.

The game will result in either a draw or a win for the intelligent piece of paper.

Slide 6

The learners will now know that the piece of paper had instructions written on one side, which were followed through by the learner who volunteered. The instructions can be described as an algorithm.

The algorithm could have been followed through by a computer if it had been written as a computer program.

Ask the class, "Does that make the piece of paper intelligent?"

Take answers from the learners. **Someone might answer that it is not the paper that is intelligent, it is the person who wrote the instructions.**

If no one gives this answer, pose it as a question to the learners: "Is it the person who wrote the instructions that is intelligent?" (You are welcome to take the credit and thank them for saying that you are intelligent.) The person who wrote the algorithm is intelligent as they have learned about IF/THEN rules and are able to apply them to create an algorithm that cannot lose in tic-tac-toe. They can also apply those IF/THEN rules to solve other problems.

Now you can congratulate the learners who thought they were more intelligent than the piece of paper to begin with, as now the class has learned that the piece of paper is not intelligent.

Ask the class, "If a computer follows an algorithm written by a human, does that make a computer intelligent?"

The answer should be that it does not. It is not learning or handling new situations, it is simply following IF/THEN rules.

10 mins

Activity 2: Rule-based vs data-driven systems

Slide 7

Explain that rule-based systems have been used for years to help us solve problems like large calculations, creating simulations and medical diagnosis tools.

The development of AI technologies has always been restricted by the technology available at the time, and the development of rule-based systems was limited by the amount of processing power available. There are also some problems that are too complex to write rules for. For example, researchers attempted to build systems for a range of specialist applications (for instance, in the areas of medicine, law, and

engineering), but they were not able to due to the number and complexity of the rules that these systems would have required.

As technology has become more advanced, a new approach has emerged called machine learning. This approach is used for almost **all modern AI systems**.

Slide 8

Explain that the learners are now going to hear about data-driven systems and machine learning from **Jessica Yung**, a research engineer at Google DeepMind.

Read the three questions on the slide and ask learners to write down answers to them as they hear them.

Play the video (3 minutes).

Ask for learners' answers to the questions.

Educator guidance

The correct answers are as follows:

- **Are AI and machine learning the same thing?** No, machine learning is **one technique** among many used to create systems that appear intelligent. A lot of modern AI systems use machine learning as it is very helpful for solving complex problems.
- **When should you use machine learning?** When the answer to the problem is **uncertain** and you need to use a lot of data to answer it.
- **What does Jessica use machine learning for?** Jessica works on YouTube autochapters. Her team makes machine learning models that analyse YouTube videos and split them into relevant chapters.

Slide 9

Ask learners to hold up the amount of fingers that correspond with the answer that they believe is correct. Use the animation to make the incorrect answer disappear. Do the same for the second question.

This allows you to assess learners' understanding of the video's content.

Slide 10

Use the slide to recap the descriptions of rule-based and data-driven approaches for the learners:

- A rule-based approach is what we would traditionally associate with programming, where an algorithm is followed to solve a computational problem
- A data-driven approach is a way of building artificial intelligence systems using statistics from vast quantities of data, instead of by writing out the rules in a program

Slide 11

The numbered boxes are animated and will appear one by one. Use these to describe the term 'machine learning' in more detail and link it to the concept of a model.

Educator guidance

A model is a representation of the real world. It is not a physical replica like a model car, but rather a set of rules, algorithms, or statistical relationships that reflect the key elements of a phenomenon or a system.

You can give the example of a weather model. A weather model is not a physical representation of the weather, but a mathematical representation of future weather conditions. It uses vast amounts of historical weather data (temperature, pressure, humidity, wind speed, etc.) to detect patterns and relationships, then uses these patterns to predict future weather conditions.

Explain that data-driven models are used to solve problems, and make the link to AI by explaining that AI chatbot applications use conversational models that are created using vast quantities of data.

Slide 12 This slide reinforces the key points from the previous slide. Use it to consolidate what you have just taught.

Slide 13 Show this slide in presenter mode. Ask the learners which box they believe should be labelled with a 1. Drag the number across the correct answer. Do this for 2 and 3.

Educator Guidance

Correct answer:

- 1: A model is trained with ML using large amounts of text data
- 2: Patterns in data are identified to create a model
- 3: The model uses patterns to generate text

15 min

Activity 3: Solving problems with machine learning

Slide 14 Ask learners what they would need to create a machine learning model to generate images of hands in different positions, from people of different genders, ages, and skin tones. **They would need data, specifically images of many different hands.**

Slide 15 View the slide in presenter mode. Again ask the learners which of the boxes should be labelled with 1, 2 and 3. Drag the numbers over to the correct answer.

Educator Guidance

Correct answer:

- 1: A model is trained with ML using many images of hands
- 2: Patterns in the data are identified to create a model
- 3: The model uses patterns to generate different images of hands

Slide 16 Explain that a data-driven approach can be used, as the model would use patterns identified in training data, not rules.

Slide 17 Show learners the output of a model that has been prompted to generate images of hands. Note that it has generated images of hands in lots of different positions.

Ask them if they notice anything about the hands.

Note that some have more or fewer than five fingers and some are in impossible positions. All hands have a similar skin tone.

Ask how learners would approach fixing this model so that it generates images that more accurately represent a range of hands.

Slide 18

Explain that to improve a data-driven system, you adjust the **quality** and **quantity** of the data, as you cannot change the inner workings of a model in the same way you could for a program.

Educator guidance

When talking about the data used to create a machine learning model, we use the terms 'quality' and 'quantity', which you may wish to explain to learners:

- **Quality:** Is the data good enough to accurately represent the thing you are modelling? Removing unclear data – for example, images that are not well lit, are blurry, or have too many other objects in them – from the dataset will make a model more accurate.
- **Quantity:** Is there enough data to represent all possible options? For example, if a model is making errors when generating images of hands holding apples, more examples might help improve it.

Slide 19

Use this slide to recap the differences between rule-based and data-driven systems, the types of problems they can be used for, and how to fix errors in them.

Ask learners to vote on whether you would use a rule-based or data-driven system to generate images of different hands.

To check their learning, ask learners to write one or two sentences to explain why you would use a data-driven system.

Slide 20

Use the animations on this slide to reveal two correct answers.

Your learners may not have used the same words as those shown on screen. They should, however, show a general understanding that writing rules for all of the possible ways that hands can be drawn would be very difficult.

Slide 21

Tell learners that they need to identify whether they would use a rule-based or data-driven approach for each scenario on the slide.

Use the animations to show learners the correct answers.

Educator guidance

These questions allow learners to review key concepts about machine learning, and allow you to check learners' understanding.

To involve the whole class, you could ask learners to give you a thumbs up if they think a data-driven approach is most suitable and a thumbs down if they think a rule-based approach is most suitable.

To address any misconceptions or ensure that learners are not guessing, you could ask a learner to explain why they have given a particular answer.

15 mins

Optional activity: Emoji predictor

Educator guidance

This is an optional extension activity. If you do not want to include it, please skip slides 20–23.

Slide 22 Optional Explain to the class that they will be training a machine learning model that matches emojis to messages.

Slide 23 Optional Show the learners the training data and explain that their model will predict an emoji based on the contents of a message.

Each column contains messages that have been assigned to the emoji at the top.

Explain that the learners are going to act as the machine learning model, and it is time to train the model.

Give the learners 5 minutes to read the messages and look for patterns in each column.

Remind them that a machine learning model cannot feel; they need to only use the words and other details about the text to predict an emoji, not their feelings about a message.

Educator guidance

The **patterns** the learners look for must relate to the **text** of the messages and not the feelings they evoke.

Instruct them to think about things like:

- The length of the messages
- Common words to start or end the messages
- Repeated words
- The format of the messages (for example, do they have two clauses joined by “and”?)

For example:

- A lot of the messages assigned to the ‘happy’ emoji start with “We”
- A lot of the messages for ‘sad’ have two clauses connected by “and”
- A lot of the messages for ‘angry’ are longer than the messages for the other two emojis

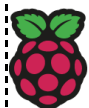
Slide 24 Optional **Testing**

Once the class has agreed on the patterns for each emoji, it is time to test the model.

Explain that the learners will use the test cards, which are the pink cards provided with the instructions on the right-hand side of the slide.

Remind the class that they can only use the text in the training examples to make their predictions, not their emotional interpretation of the test messages. Tell them you will ask for a justification to support their predictions. If they cannot match the text or patterns, they have to make a guess.

Give the class 5 minutes to examine the test messages and then ask for their predictions. Ask them to justify their predictions with the patterns discussed earlier or other details based on the training data.



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Educator guidance

After each prediction, ask the class these three questions:

- "Was it easy or hard to decide?"
- "Did we have enough examples that are similar?"
- "Could the model be wrong here? Why?"

The aim is for them to reflect on the quality of their predictions, and the effect that the training data has on their 'confidence' in the predictions they are making.

Below are some possible answers. These are not the only correct answers, as learners may have identified other patterns.

- "We waited for ages, and still didn't get tickets" starts with "We", so could be matched with 😊
- "They liked my painting!" starts with "They", so could be matched with 😡
- "My project got moved and now it's ruined" starts with "My" and includes "and", so could be matched with 😞
- "We had to wait a long time" starts with "We", so could be matched with 😊

Slide 25 Optional Evaluate

Use the questions on the slide to evaluate the emoji prediction model.

Was the model accurate? How often did the 'predictions' match the emoji the class would assign to that message? See if you can produce a percentage.

Why did the model predict the wrong emoji? Encourage the class to reflect on the **quality** and **quantity** of the training data.

Finally, ask the class **how they could improve the model**, and take suggestions for the types of data they would add.

5 mins

Exit ticket

Slide 26

Ask learners to complete the sentences using some of the words on the slide.

Slide 27

Reveal the completed sentences.

Slide 28

Review the learning objectives and look ahead to the next lesson.